

Troubleshooting Kubernetes Monitoring

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Debugging Dynatrace Monitoring in Kubernetes

When monitoring doesn't work as expected, systematic troubleshooting is essential. This notebook covers common issues, diagnostic procedures, and resolution steps for Dynatrace Kubernetes monitoring.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS/Managed access
kubectI	Configured for your cluster
Permissions	Cluster admin or debug access
Knowledge	K8S-01 through K8S-03

1. Troubleshooting Methodology

Systematic Approach

 Troubleshooting Methodology

Common Symptom Categories

Symptom	Likely Component	Start Here
No cluster data	ActiveGate	Section 4
No node/pod data	OneAgent	Section 3
No app traces	Injection	Section 5
Partial data	Data collection	Section 6
Operator errors	Operator	Section 2

First Checks

```
# DynaKube status
kubectl -n dynatrace get dynakube

# All Dynatrace pods
kubectl -n dynatrace get pods

# Recent events
kubectl -n dynatrace get events --sort-by='.lastTimestamp' | tail -20
```

2. Operator Issues

Operator Not Running

Symptoms:

- DynaKube CR shows no status
- OneAgent pods not created

Diagnostic:

```
# Check operator deployment
kubectl -n dynatrace get deployment dynatrace-operator

# Check operator logs
kubectl -n dynatrace logs -l app.kubernetes.io/name=dynatrace-operator --
tail=100
```

Common Causes:

Cause	Resolution
Missing CRDs	Reinstall operator with Helm
RBAC issues	Check ServiceAccount permissions
Resource limits	Increase operator resources
Webhook failure	Check webhook certificates

DynaKube Stuck in Deploying

Diagnostic:

```
# Describe DynaKube
kubectl -n dynatrace describe dynakube dynakube

# Check conditions
kubectl -n dynatrace get dynakube -o jsonpath='{.items[*].status.conditions}'
| jq .
```

Common Causes:

Cause	Resolution
Invalid API token	Regenerate and update secret
Network policy blocking	Allow egress to Dynatrace
Image pull failure	Check pull secrets

3. OneAgent Issues

OneAgent Pods Not Starting

Diagnostic:

```
# Check DaemonSet
kubectl -n dynatrace get daemonset -l app.kubernetes.io/component=oneagent

# Check pods
kubectl -n dynatrace get pods -l app.kubernetes.io/component=oneagent -o wide

# Describe failing pod
kubectl -n dynatrace describe pod &lt;oneagent-pod-name>;
```

Common Causes:

Cause	Resolution
Tolerations missing	Add tolerations to DynaKube
Node selector mismatch	Update node selectors
Resource unavailable	Check node resources
Security context denied	Configure PSP/PSS

OneAgent Not Reporting

Diagnostic:

```
# Check OneAgent logs
kubectl -n dynatrace logs <oneagent-pod> -c oneagent --tail=100

# Check connectivity
kubectl -n dynatrace exec <oneagent-pod> -c oneagent -- curl -v
https://<tenant>.live.dynatrace.com/api/v1/deployment/installer/agent/c
onnectioninfo
```

Common Causes:

Cause	Resolution
Firewall blocking	Allow egress port 443
Proxy misconfiguration	Set proxy in DynaKube
Certificate issues	Check custom CA config

4. ActiveGate Issues

ActiveGate Not Starting

Diagnostic:

```
# Check StatefulSet
kubectl -n dynatrace get statefulset -l
app.kubernetes.io/component=activegate

# Check pods
kubectl -n dynatrace get pods -l app.kubernetes.io/component=activegate

# Check logs
kubectl -n dynatrace logs &lt;activegate-pod> --tail=100
```

Common Causes:

Cause	Resolution
Memory too low	Increase resources in DynaKube
PVC issues	Check storage class
Token invalid	Regenerate API token

No Kubernetes Metrics

Diagnostic:

```
# Check capabilities
kubectl -n dynatrace get dynakube -o
jsonpath='{.items[*].spec.activeGate.capabilities}'

# Should include: kubernetes-monitoring
```

Resolution: Ensure `kubernetes-monitoring` capability is enabled:

```
spec:
  activeGate:
    capabilities:
      - kubernetes-monitoring
      - routing
```

5. Injection Issues

Pods Not Getting Injected

Diagnostic:

```
# Check namespace labels
kubectl get namespace <namespace> -o jsonpath='{.metadata.labels}'

# Check pod annotations
kubectl get pod <pod> -o jsonpath='{.metadata.annotations}'

# Check webhook
kubectl get mutatingwebhookconfigurations
```

Common Causes:

Cause	Resolution
Namespace not selected	Add matching labels
Pod has opt-out annotation	Remove annotation
Webhook not registered	Restart operator
CSI driver not mounted	Enable CSI in DynaKube

Check Injection Status

```
# Check if init container present
kubectl get pod <pod> -o jsonpath='{.spec.initContainers[*].name}'

# Check environment variables
kubectl exec <pod> -c <container> -- env | grep DT_
```

Namespace Selector Debugging

If using `namespaceSelector` in DynaKube:

```
# DynaKube selector
kubectl -n dynatrace get dynakube -o
jsonpath='{.items[*].spec.namespaceSelector}'

# Label a namespace for injection
kubectl label namespace &lt;namespace> dynatrace-injection=enabled
```

6. Data Collection Issues

Missing Metrics

Diagnostic Steps:

1. Verify entity exists in Dynatrace
2. Check metric availability for entity type
3. Verify time range includes data
4. Check for filtering issues

```
// Verify Kubernetes entities are being discovered
fetch dt.entity.kubernetes_cluster
| fields entity.name, lifetime
| sort lifetime desc
```

```
// Check if container metrics are flowing
timeseries cpu = avg(dt.kubernetes.container.cpu_usage), from:-1h
| limit 1
```

```
// Check if logs are being ingested
fetch logs, from: now() - 1h
| filter isNotNull(k8s.namespace.name)
| summarize logCount = count()
```

Missing Logs

Diagnostic:

```
# Check if log analytics is enabled
kubectl -n dynatrace get dynakube -o yaml | grep -A5 "env:"

# Verify OneAgent log collection
kubectl -n dynatrace exec <oneagent-pod> -c oneagent -- cat
/var/lib/dynatrace/oneagent/log/oneagent.log | tail -50
```

Common Causes:

Cause	Resolution
Log ingest disabled	Enable in tenant settings
Log volume too high	Configure sampling/filtering
Log format unsupported	Configure custom parsing

7. Connectivity Issues

Testing Connectivity

```
# From OneAgent pod
kubectl -n dynatrace exec <oneagent-pod> -c oneagent -- \
  curl -v https://<tenant>.live.dynatrace.com/api/v1/time

# From ActiveGate pod
kubectl -n dynatrace exec <activegate-pod> -- \
  curl -v https://<tenant>.live.dynatrace.com/api/v1/time
```

Network Policy Issues

```
# Allow Dynatrace egress
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-dynatrace-egress
  namespace: dynatrace
spec:
  podSelector: {}
  policyTypes:
    - Egress
  egress:
    - to:
        - ipBlock:
            cidr: 0.0.0.0/0
      ports:
        - protocol: TCP
          port: 443
```

Proxy Configuration

```
spec:
  proxy:
    value: https://proxy.example.com:8080
  # Or use a secret
  # proxy:
  #   valueFrom:
  #     secretKeyRef:
  #       name: proxy-secret
  #       key: proxy
```

8. Diagnostic Commands

Quick Health Check

```
#!/bin/bash
# dynatrace-health-check.sh

echo "=== DynaKube Status ==="
kubectl -n dynatrace get dynakube

echo "\n=== Dynatrace Pods ==="
kubectl -n dynatrace get pods

echo "\n=== OneAgent DaemonSet ==="
kubectl -n dynatrace get daemonset -l app.kubernetes.io/component=oneagent

echo "\n=== ActiveGate StatefulSet ==="
kubectl -n dynatrace get statefulset -l app.kubernetes.io/component=activegate

echo "\n=== Recent Events ==="
kubectl -n dynatrace get events --sort-by='.lastTimestamp' | tail -10
```

Collect Support Bundle

```
# Create support bundle
mkdir dynatrace-debug
cd dynatrace-debug

# DynaKube
kubectl -n dynatrace get dynakube -o yaml &gt; dynakube.yaml

# Pods
kubectl -n dynatrace get pods -o yaml &gt; pods.yaml

# Events
kubectl -n dynatrace get events &gt; events.txt

# Operator logs
kubectl -n dynatrace logs -l app.kubernetes.io/name=dynatrace-operator &gt; operator.log

# OneAgent logs (one pod)
kubectl -n dynatrace logs $(kubectl -n dynatrace get pods -l app.kubernetes.io/component=oneagent -o jsonpath='{.items[0].metadata.name}') -c oneagent &gt; oneagent.log

# Package
cd ..
tar -czf dynatrace-debug.tar.gz dynatrace-debug/
```

Common kubectl Commands

Command	Purpose
<code>kubectl -n dynatrace get all</code>	All resources
<code>kubectl -n dynatrace describe dynakube</code>	DynaKube details
<code>kubectl -n dynatrace logs <pod></code>	Pod logs
<code>kubectl -n dynatrace exec -it <pod> -- sh</code>	Shell access
<code>kubectl -n dynatrace port-forward <pod> 9999:9999</code>	Local port forward

Summary

In this notebook, you learned:

- Systematic troubleshooting methodology
 - Operator debugging and common issues
 - OneAgent troubleshooting steps
 - ActiveGate diagnostics
 - Code injection debugging
 - Data collection issue resolution
 - Connectivity testing and network policy configuration
 - Diagnostic commands for support bundle collection
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References

- [Dynatrace Operator Troubleshooting](#)
 - [OneAgent Troubleshooting](#)
 - [Kubernetes Monitoring FAQ](#)
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